

ATmega32A Module

Timers (Normal, CTC and Fast PWM)

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1 Theory

A hardware timer is a counter. It counts clock ticks automatically, and your program does not need to do anything while it counts. The ATmega32A has three timers:

Timer	Size	Counts	Prescalers	OC pin(s)
Timer0	8-bit	0–255	1, 8, 64, 256, 1024	PB3 (LED D3)
Timer1	16-bit	0–65535	1, 8, 64, 256, 1024	PD5 (A), PD4 (B)
Timer2	8-bit	0–255	1, 8, 32, 64, 128, 256, 1024	PD7

Prescaler. The prescaler N divides the CPU clock before it goes to the counter. So one count takes $t_{\text{count}} = N/F_{\text{CPU}}$. Example: with 8 MHz and $N = 1024$, one count takes $128 \mu\text{s}$.

The three modes.

- **Normal** (TIMER_NORMAL): The timer counts up to the maximum (255, or 65535 for Timer1). Then it goes back to 0. This is called an *overflow*, and the timer sets the overflow flag TOV. For a shorter time, you give the counter a start value (a *preload*):

$$t = (256 - \text{preload}) \cdot \frac{N}{F_{\text{CPU}}} \quad (\text{use } 65536 \text{ for Timer1})$$

- **CTC** (TIMER_CTC, *Clear Timer on Compare*): The timer counts up to the compare value OCR. Then it sets the compare flag OCF and starts again from 0. You do not need a preload:

$$t = (\text{OCR} + 1) \cdot \frac{N}{F_{\text{CPU}}}$$

On every compare match, the timer can also toggle, set or clear its OC pin. This gives a square wave, and the CPU does not need to do anything.

- **Fast PWM** (TIMER_FAST_PWM): The timer counts from 0 to 255 again and again. The OC pin changes at 0 and at the compare value. The result is a PWM signal. OCR sets the duty cycle:

$$F_{\text{PWM}} = \frac{F_{\text{CPU}}}{N \cdot 256} \quad \text{Duty cycle (active part)} = \frac{\text{OCR} + 1}{256}$$

2 How to use the driver

The driver gives you three objects: Timer0, Timer1 and Timer2. You start a timer with one function call:

```
Timer0.begin(mode, prescaler);           // start the timer
Timer0.begin(mode, prescaler, pinMode);   // start the timer and use the OC pin
```

The options

Argument	Value	Meaning
mode	TIMER_NORMAL	Count from 0 to the maximum, overflow, start again from 0
	TIMER_CTC	Count from 0 to the compare value, start again from 0
	TIMER_FAST_PWM	Count from 0 to 255, the OC pin gives a PWM signal
prescaler	1, 8, 64, 256, 1024	Write the number you want to divide by. Timer2 can also use 32 and 128.
	EXT_FALLING, EXT_RISING	Only Timer0 and Timer1: count pulses on pin T0 = PB0 or T1 = PB1 (counter mode)
pinMode	OC_OFF (default)	The OC pin is a normal GPIO pin
	<i>with TIMER_NORMAL or TIMER_CTC:</i>	
	OC_TOGGLE	Toggle the OC pin on every compare match
	OC_CLEAR	Set the OC pin LOW on a compare match
	OC_SET	Set the OC pin HIGH on a compare match
	<i>with TIMER_FAST_PWM:</i>	
	PWM_NON_INVERTING	HIGH from 0 to the compare value, then LOW
	PWM_INVERTING	LOW from 0 to the compare value, then HIGH. Use this for the active-low LEDs on the board.

If `pinMode` is not `OC_OFF`, the driver makes the OC pin an output for you. While the timer controls this pin, `GPIO.write()` has no effect on it.

Functions

Function	What it does
<code>Timer0.begin(mode, prescaler, pinMode)</code>	Set up and start the timer (see the options above)
<code>Timer0.setCount(value)</code>	Write a start value to the counter (example: 156 gives 100 counts in Normal mode)
<code>Timer0.getCount()</code>	Read the counter (0–255; Timer1: 0–65535)
<code>Timer0.setCompare(value)</code>	Set the compare value. CTC: the timer counts up to this value. PWM: the level, 0–255
<code>Timer0.setDutyCycle(percent)</code>	Only PWM: how many percent (0–100) of the time the pin is <i>active</i> . Active is HIGH for <code>PWM_NON_INVERTING</code> and LOW for <code>PWM_INVERTING</code> . 0% and 100% are exact.
<code>Timer0.overflowed()</code>	Returns <code>true</code> one time for each overflow. The flag is cleared for you
<code>Timer0.compareMatched()</code>	Returns <code>true</code> one time for each compare match. The flag is cleared for you
<code>Timer0.stop()</code> / <code>Timer0.start()</code>	Stop / continue counting. The settings do not change
<code>Timer0.onOverflow(function)</code>	Call <code>function</code> on every overflow (interrupt, see the interrupt lecture)
<code>Timer0.onCompareMatch(function)</code>	Call <code>function</code> on every compare match (interrupt, see the interrupt lecture)
<i>Only Timer1 (it has two OC pins: A = PD5, B = PD4):</i>	
<code>Timer1.begin(mode, N, pinModeA, pinModeB)</code>	Channel B has its own <code>pinMode</code>
<code>Timer1.setCompareA/B(value)</code>	16-bit compare values. <code>setCompare()</code> is channel A. In CTC mode the timer counts up to A
<code>Timer1.setDutyCycleA/B(percent)</code>	PWM duty cycle for each channel. <code>setDutyCycle()</code> is channel A
<code>Timer1.compareMatchedA/B()</code>	Compare flag for each channel. <code>compareMatched()</code> is channel A
<code>Timer1.setTop(top)</code>	Length of the PWM period in counts (default 255): $T_{\text{PWM}} = (\text{top} + 1) \cdot N / F_{\text{CPU}}$
<i>Timer2 (8-bit, OC pin = PD7):</i>	
<code>Timer2.begin(mode, N, pinMode)</code>	All functions are the same as for <code>Timer0</code> . Only the prescaler list is different: 1, 8, 32, 64, 128, 256, 1024 (no <code>EXT_FALLING</code> / <code>EXT_RISING</code>)

Waiting for a flag (polling). `overflowed()` and `compareMatched()` return `true` when the event happened after your last call. So `while (!Timer0.overflowed());` waits until the next overflow. You do not need to write 1 to TIFR.

Interrupts. `onOverflow()` and `onCompareMatch()` turn on the interrupt and call `sei()` for you. Your function then runs automatically. Keep this function short. If it changes a variable that `main()` also uses, declare the variable as `volatile`. To turn the interrupt off again, pass `nullptr`. We come back to interrupts in the interrupt lecture.

Driver functions and registers

In the lecture slides we set the registers by hand. Each driver function does one of these steps for you:

Driver function	Registers (Timer0)	Bits
<code>Timer0.begin(TIMER_NORMAL, 1024)</code>	<code>TCCR0 = 0x05;</code>	CS02 CS00
<code>Timer0.begin(TIMER_CTC, 1024, OC_TOGGLE)</code>	<code>TCCR0 = 0x1D;</code>	WGM01 COM00 CS02 CS00
	<code>DDRB = 1<<PB3;</code>	OC0 pin as output
<code>Timer0.begin(TIMER_FAST_PWM, 1, PWM_INVERTING)</code>	<code>TCCR0 = 0x79;</code>	WGM00 WGM01 COM01 COM00 CS00
<code>Timer0.setCount(156)</code>	<code>TCNT0 = 156;</code>	
<code>Timer0.setCompare(x)</code>	<code>OCR0 = x;</code>	
<code>Timer0.overflowed()</code>	<code>TIFR & (1<<TOV0)</code>	then <code>TIFR = 1<<TOV0;</code>
<code>Timer0.compareMatched()</code>	<code>TIFR & (1<<OCF0)</code>	then <code>TIFR = 1<<OCF0;</code>

For `Timer2` the registers are `TCCR2`, `TCNT2`, `OCR2` and the flags `TOV2`, `OCF2`, and the bits have the same places as in `TCCR0`. Only the prescaler bits are different. Example: `Timer2.begin(TIMER_CTC, 32)` gives `TCCR2 = 0x0B` (`WGM21 CS21 CS20`). With `Timer0`, the same bits `CS01 CS00` mean prescaler 64. The driver chooses the right bits for you.

Important: F_CPU and the fuses. All timing formulas use the *real* clock of the chip. `make` and `make timer` always set the fuses for the external 8 MHz crystal. To use another clock: (1) set the fuses, for example `make int-1mhz`; (2) upload your program without changing the fuses: `make build flash SRC=Task.cpp`; (3) change `#define F_CPU` to the new clock, so that `_delay_ms()` is still correct.

3 Example: blinking LED and PWM, both done by the timers

Board hardware. All parts are already on the development board:

- **8 LEDs (D0–D7) on PORTB.** They are active-low: LOW = ON. LED D3 is on PB3 = OC0, the output pin of Timer0.
- **Seven-segment display,** also on PORTB (it uses the same pins as the LEDs).
- **8 DIP switches (S0–S7) on PORTC.** A closed switch reads LOW. Use INPUT_PULLUP.
- **Push buttons:** S11 on PD2 and S10 on PD6. A pressed button reads LOW.

Build and upload. `make timer`

```
/*
 * Timer Example -- one timer per mode, all running at the same time
 * Build & flash: make timer
 *
 * LED D0 (PB0): blinks every 0.5 s -- Timer1, CTC mode, flag polling
 * LED D3 (PB3): half brightness -- Timer0, Fast PWM on its OC0 pin
 *
 * Timer1 CTC period = prescaler * (compare + 1) / F_CPU
 *                   = 256 * (15624 + 1) / 8 MHz = 0.5 s
 */

#define F_CPU 8000000UL
#include "../gpio/gpio.hpp"
#include "timer.hpp"

int main() {
    GPIO.setDirection(PORTB, PB0, OUTPUT); // LED D0 (active-low)

    // Timer0: Fast PWM, prescaler 64 -> 8 MHz / (64 * 256) = 488 Hz, no flicker.
    // PWM_INVERTING because the LEDs are active-low (duty cycle = brightness).
    Timer0.begin(TIMER_FAST_PWM, 64, PWM_INVERTING); // PB3 becomes an output
    Timer0.setDutyCycle(50);

    // Timer1: CTC, restarts every 15625 counts = 0.5 s
    Timer1.begin(TIMER_CTC, 256);
    Timer1.setCompare(15624);

    while (true) {
        if (Timer1.compareMatched()) { // true once every 0.5 s
            GPIO.toggle(PORTB, PB0);
        }
        // The CPU is free here -- the PWM on D3 needs no code at all.
    }
}
```

4 Laboratory Tasks

For each task, make a new `.cpp` file in the main project folder (for example `Task1.cpp`). Start the file with `#define F_CPU`. Then include the drivers you need: `drivers/gpio/gpio.hpp`, `drivers/timer/timer.hpp`, `drivers/common/sevenseg.hpp`. Upload it with `make SRC=Task1.cpp`.

Task 0: 1 MHz or 8 MHz? (10 min)

Show the numbers 0 to 9 on the seven-segment display. Change the number every second. Use `SevenSeg(digit)` and `_delay_ms(1000)`, and write `#define F_CPU 8000000UL` at the top of the file.

1. Upload with `make SRC=Task0.cpp`. This also sets the chip to the external 8 MHz crystal. Check with a stopwatch: is one step one second?
2. Now change the chip to the internal 1 MHz clock with `make int-1mhz`. Do not upload the program again.

What is the difference? How long is one “second” now? Explain why.

Task 1: Timer Normal Mode — Counter on the LEDs (30 min)

Count in binary on the 8 LEDs. Add 1 every **100 ms**. Use `Timer0` in `TIMER_NORMAL` mode with prescaler 1024 for the timing (external 8 MHz crystal). The LEDs are active-low, so write the inverted value: `GPIO.write(PORTB, ~count)`.

One `Timer0` overflow is too short for 100 ms, so use 10 overflows of 10 ms. Calculate the preload value for 10 ms. In the main loop, do these two steps 10 times:

1. write the preload with `Timer0.setCount()`;
2. wait with `while (!Timer0.overflowed());`.

After that, add 1 to the counter.

Task 2: Timer CTC Mode — Blinking in Hardware (20 min)

The timer blinks LED **D3** (PB3 = OC0) without help from the CPU. The DIP switches on `PORTC` set the blink speed.

Start `Timer0` in `TIMER_CTC` mode with prescaler 1024 and `OC_TOGGLE`. The main loop only does one thing: it reads the switches (`GPIO.read(PINC)`) and writes the value with `Timer0.setCompare()`. You do not need to check any flag. At 8 MHz, calculate the slowest possible blink frequency. What do you see on D3? Explain why.

Task 3: Timer Fast PWM — LED Brightness (25 min)

Change the brightness of LED **D3** (OC0) with PWM from `Timer0`.

- **Task 3.1 (Switches → brightness):** Start `Timer0` in `TIMER_FAST_PWM` mode with `PWM_INVERTING`, because the LED is active-low. Choose the prescaler so that $F_{\text{PWM}} \approx 4 \text{ kHz}$ at 8 MHz. In the loop, write `Timer0.setCompare(GPIO.read(PINC))`.
- **Task 3.2 (Buttons, in percent):** Use the push buttons instead of the switches. S11 (PD2) makes the LED 10 % brighter. S10 (PD6) makes it 10 % darker. Keep the value between 0 and 100 %. Use `Timer0.setDutyCycle(percent)`. One press must change the value only one time, so wait until the button is released.

- **Task 3.3 (Breathing LED with two timers):** LED D3 should slowly become brighter from 0 % to 100 %, and then slowly darker from 100 % to 0 %. This repeats all the time.

Task 3.3 step by step. You need two timers, and each timer has one job:

- **Timer0 makes the brightness.** It creates the PWM signal on D3, exactly as in Task 3.1 (Fast PWM, about 4 kHz, PWM_INVERTING).
- **Timer1 is a clock that “ticks” every 10 ms.** Use TIMER_CTC mode. Every 10 ms there is a compare match, and `Timer1.compareMatched()` returns `true` one time.

Calculate the Timer1 compare value with $t = (\text{OCR} + 1) \cdot N / F_{\text{CPU}}$. Hint: with prescaler $N = 64$ at 8 MHz, one count takes $8 \mu\text{s}$. How many counts are 10 ms?

On every tick, change the brightness by 1 %. Use two variables: `percent` (the brightness now, 0–100) and `step` (+1 for brighter, -1 for darker). When `percent` reaches 100, change `step` to -1. When it reaches 0, change `step` to +1. One full “breath” then takes $200 \times 10 \text{ ms} = 2 \text{ s}$.

```
Timer0.begin(TIMER_FAST_PWM, ??, PWM_INVERTING); // brightness (PWM on D3)
Timer1.begin(TIMER_CTC, 64);                      // tick every 10 ms
Timer1.setCompare(??);

int8_t percent = 0;
int8_t step = +1;

while (true) {
  if (Timer1.compareMatched()) { // true once every 10 ms
    percent = percent + step;
    if (percent == ??) step = ??; // top reached: go darker
    if (percent == ??) step = ??; // bottom reached: go brighter
    Timer0.setDutyCycle(percent);
  }
}
```

Why is this better than `_delay_ms(10)`? The CPU never waits, so the loop can do other work at the same time (for example, read the buttons).

Expected result. You can watch a recording of the finished task on the board in the repository (click the link):

`drivers/timer/task3_3.gif`